

Docker Image Not Found SOP

Objective

Resolve image pull failures when Docker cannot find or access the required image from a local cache or remote registry.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
- 2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
- 3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
- 4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
- 5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T040
Issue	Docker image not found
Category	Deployment
Service	Docker
Severity	Medium
Status	Resolved
Description	Image pull fails
Expected Resolution	Check image name and registry access

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L010	5/1/2026 10:05	ERROR	Failed to pull image	DOCKER_PULL
L030	5/1/2026 10:15	WARNING	Failed to pull image	DOCKER_PULL
L050	5/1/2026 10:25	ERROR	Failed to pull image	DOCKER_PULL

Procedure

- 1. Confirm image name, tag, registry, namespace, and deployment source.
- 2. Run docker pull manually to reproduce the failure and capture the error message.
- 3. Check for spelling errors, missing tag, private registry namespace, or deleted image.
- 4. Validate registry login, credentials, network access, and proxy configuration.
- 5. Confirm the image exists in the registry and the tag was pushed by CI/CD.
- 6. Update the image reference, authenticate to registry, or rebuild and push the missing image.
- 7. Pull the corrected image and run a smoke test.
- 8. Update deployment configuration with the verified image reference.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Image not found	Wrong image name or tag	Correct image reference
Pull denied	Registry auth missing	Run docker login or update credentials
Tag missing	CI did not push image	Rebuild and push image
Registry timeout	Network or proxy issue	Check connectivity and proxy settings

Best Practices

1. Use immutable image tags and registry namespaces.
2. Validate image push in CI/CD.
3. Avoid relying on latest tag.
4. Keep registry credentials rotated and tested.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook